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Problem

CONTEXT IS THE BOTTLENECK

- Real tasks combine long document collections, hours of media and details separated by millions of tokens.
- Short windows force retrieval pipelines and prevent one model from jointly reasoning over all supplied evidence.

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Recall & learning

MORE THAN 99% UP TO 10M

- Needle-in-a-haystack retrieval remains near perfect across modalities and exceeds 99% up to at least 10M tokens.
- Given only a grammar manual, the model learns in context to translate the low-resource language Kalamang.

2

Model family

PRO AND FLASH

- Gemini 1.5 uses a compute-efficient multimodal mixture-of-experts architecture derived from Gemini 1.0.
- Pro prioritizes capability; Flash is a lighter serving-oriented variant with small quality regression.

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Evidence

LONG-INPUT STATE OF THE ART

- Gemini 1.5 advances long-document QA, long-video QA and long-context ASR while matching or surpassing Gemini 1.0 Ultra broadly.
- Professional collaborations across ten job categories report 26% to 75% time savings.

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Long multimodal context

DOCUMENTS, VIDEO AND AUDIO

- The models ingest long text, code repositories, many images, hours of video and audio in one context.
- A standard million-token window is stress-tested to at least ten million tokens.

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Meaning & limits

- The work turns long context into a unified workspace for multimodal retrieval, analysis and rapid in-context adaptation.
- Needle recall is easier than deep synthesis; very long prompts raise latency, cost and privacy risks, and system details remain proprietary.

GEMINI 1.5

Gemini 1.5: Millions of Tokens

Gemini Team | arXiv:2403.05530v5

Extend a compute-efficient multimodal model family to recall and reason over millions of tokens spanning documents, video and audio.

long context

multimodal MoE

Pro + Flash